The same reasoning holds, of course, if one wants to probe the pit even more deeply to determine its atomic and subatomic structure. Half-millimeter pellets will not provide any useful information; they are clearly too big to have any sensitivity to structure on atomic scales. This is why particle accelerators use protons or electrons as probes, since their small size makes them much better suited to the task. On subatomic scales, where quantum concepts replace classical reasoning, the most appropriate measure of a particle's probing sensitivity is its quantum wavelength, which indicates the window of uncertainty in its position. This fact reflects our discussion of Heisenberg's uncertainty principle in Chapter 4, in which we found that the margin of error incurred when using a point particle as a probe (we focused on photon probes but the discussion applies to all other particles) is about equal to the probe particle's quantum wavelength. In somewhat looser language, the probing sensitivity of a point particle is smeared out by the jitteriness of quantum mechanics, in much the same way that the precision of a surgeon's scalpel is compromised if he or she has hands that shake. But recall that in Chapter 4 we also noted the important fact that a particle's quantum wavelength is inversely proportional to its momentum, which, roughly speaking, is its energy. And so, by increasing a point particle's energy, its quantum wavelength can be made shorter and shorter—quantum smearing can be decreased further and further—and hence we can use it to probe ever finer physical structures. Intuitively, higher-energy particles have greater penetrating power and are therefore able to probe more minute features.

In this regard, the distinction between point particles and strands of string becomes manifest. Just as was the case for plastic pellets probing the surface features of a peach pit, the string's inherent spatial extent prevents it from probing the structure of anything substantially smaller than its own size—in this case structures arising on length scales shorter than the Planck length. Somewhat more precisely, in 1988 David Gross, then of Princeton University, and his student Paul Mende showed that when quantum mechanics is taken into account, continually increasing the energy of a string does not continually increase its ability to probe finer structures, in direct contrast with what happens for a point particle. They found that when the energy of a string is increased, it is at first able to probe shorter-scale structures,